



Koneru Lakshmaiah Education Foundation

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OPERATING SYSTEMS AND SYSTEMS PROGRAMMING (25CS2104E)

2026–27, Term-I

Project Problem Statement Submission Form

Section / Team Details

Section No.: 10

Team No.: 15

Project Title: Kernel Monitor

Team Members

Roll Number	Student Name	Signature
2520090117	V Sri Vaishnavi	Vaishnavi
2520090176	Y Mohana Priya	Mohana

1. Abstract

Kernel Monitor is a Linux-based system monitoring tool designed to provide users with a clear view of important operating system and kernel-level information. Monitoring system resources and processes is essential for understanding system performance, identifying resource-intensive processes, and detecting potential system issues. The project focuses on relevant Operating Systems and System Programming concepts such as process management, CPU and memory utilization, system calls, Linux /proc filesystem, process states, and system information retrieval.

The proposed solution will be implemented in Linux using C/C++ and Linux/POSIX APIs. The tool will collect and display information such as CPU usage, memory utilization, system uptime, kernel version, running processes, process IDs, process states, and resource consumption. It will provide a simple interface for viewing system information and monitoring processes in real time. Selected process-control operations can also be incorporated using Linux signals.

The expected outcome is a lightweight and user-friendly Kernel Monitor that demonstrates how applications interact with the Linux operating system to retrieve system-level information and manage processes. The project will provide practical understanding of Linux system programming and Operating Systems concepts while helping users monitor system performance efficiently.

2. Problem Statement

Modern computer systems run multiple processes and utilize several resources such as CPU, memory, and storage simultaneously. Without proper monitoring, it can be difficult to identify resource-intensive processes, understand process states, or determine the current condition of the operating system.

The **Kernel Monitor** project aims to develop a Linux-based monitoring tool that collects and displays important system and process information in an organized manner. The project addresses Operating Systems and System Programming concepts including process management, resource management, system calls, process states, and interaction with the Linux `/proc` filesystem.

The system will provide users with real-time information about CPU and memory utilization, running processes, kernel details, and system uptime, making it easier to understand and monitor the behavior of the Linux operating system.

3. Objectives

1. **To monitor important system resources** such as CPU and memory utilization in a Linux environment.
2. **To display running process information**, including Process ID (PID), process name, process state, and resource usage.
3. **To retrieve kernel and system information** such as kernel version, system architecture, and system uptime.
4. **To demonstrate Linux Operating Systems concepts** such as process management, system calls, signals, and the `/proc` filesystem through a practical application.

4. Proposed Methodology

The Kernel Monitor will be designed and implemented as a Linux-based system monitoring application using **C/C++ and Linux/POSIX APIs**. The application will interact with the Linux operating system to collect system-level information.

The project will primarily use the **`/proc` virtual filesystem** to obtain information about CPU utilization, memory usage, running processes, system uptime, and other kernel-related information. Process-related information will be obtained by reading appropriate `/proc` entries and using Linux system interfaces.

The application will contain separate modules for **system information, CPU monitoring, memory monitoring, process monitoring, and process control**. The monitoring module will periodically collect updated information and display it to the user. Process-control operations, where implemented, will use Linux signals and appropriate system calls.

Basic flow:

Start → Collect Kernel/System Information → Collect CPU & Memory Data → Read Process Information → Display Monitoring Data → Refresh → Exit

The final application will provide a simple and organized interface through which users can observe the current state and resource usage of the Linux system.

5. Operating Systems Concepts / Linux APIs Used

(List the relevant Operating Systems concepts, Linux system calls, POSIX APIs, or mechanisms that will be used.)

OS Concept / Linux API / System Call	Purpose in the Project
Process Management	Monitor and manage running processes
Process ID (PID)	Identify individual processes
Signals	Support process termination/control operations
CPU & Memory Management	Monitor resource utilization
/proc Filesystem	Retrieve kernel, CPU, memory and process information
Process States	Display whether processes are running, sleeping, stopped, etc.

6. Individual Contribution

(Clearly specify the responsibility of each team member in the project.)

Roll Number	Student Name	Individual Responsibility
2520090117	V Sri Vaishnavi	Project coordination, system information module and integration
2520090176	Y Mohana Priya	Process monitoring and process information module

7. Tools / Platforms / Software Used

(List the programming languages, operating system, libraries, tools, and software planned for the project.)

Tool / Platform / Software	Purpose
Linux / Ubuntu	Operating environment for development and execution
C / C++	Core programming and system-level implementation
Linux System Calls	Interaction with operating system services
POSIX APIs	Process and system-level operations
Terminal / Bash	Running, testing and managing the application

8. Expected Outcome

The proposed **Kernel Monitor** is expected to provide a lightweight Linux-based tool for monitoring important system and process information. The application will display CPU utilization, memory usage, system uptime, kernel information, and details of currently running processes such as PID, process name, process state, and resource consumption.

The project is also expected to demonstrate practical implementation of Operating Systems and System Programming concepts, including process management, Linux system calls, signals, file I/O, POSIX APIs, and the /proc filesystem.

The final system will provide an easy-to-understand monitoring interface and help users observe system resource usage and process behavior. It will also give the project team practical experience in developing applications that interact directly with the Linux operating system.

9. GitHub Repository

GitHub Repository Link: <https://github.com/mohanapriya-blip/OSSP-.git>

Repository Created: _____

☐ Yes ☐ No

Faculty Name and Signature

Faculty Name: Raghupathi

Signature: _____

Remarks: _____

Faculty Use

Project Approved: _____

☐ Yes ☐ No

Date: _____

Faculty Signature: _____